

We'll Cross That Bridge When We Get There: Ancestral Trait Reconstruction with Monte Carlo Diffusion Bridge Sampling

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Ancestral Trait Reconstruction

Ancestral Trait Reconstruction (ATR) is the process of reconstructing historical trait values of unknown ancestor species, in relation to some phylogenetic tree, such as the one seen in *Figure 2* this relies on a fundamental idea; Species that are closely related are likely to have more similar trait values than those that are further apart¹. However properly accounting for this shared history, and other ecological factors has been a major struggle for evolutionary biologists for almost 50 years, as there is a significant loss in statistical convenience when working with non-independent data.

Historically this problem is avoided, by using normally distributed Stochastic Differential Equations (SDEs) such as Brownian Motion (BM) or Ornstein-Uhlenbeck Processes (OU) as the underlying model of evolution. We wish to remedy this by extending these processes to take non-independence into account. This will allow us to more accurately model evolutionary processes and include additional parameters that may be incredibly useful such as ecological or environmental factors that change between time and species. Additionally, we aim to allow for better multivariate methods to more accurately model within-species correlations

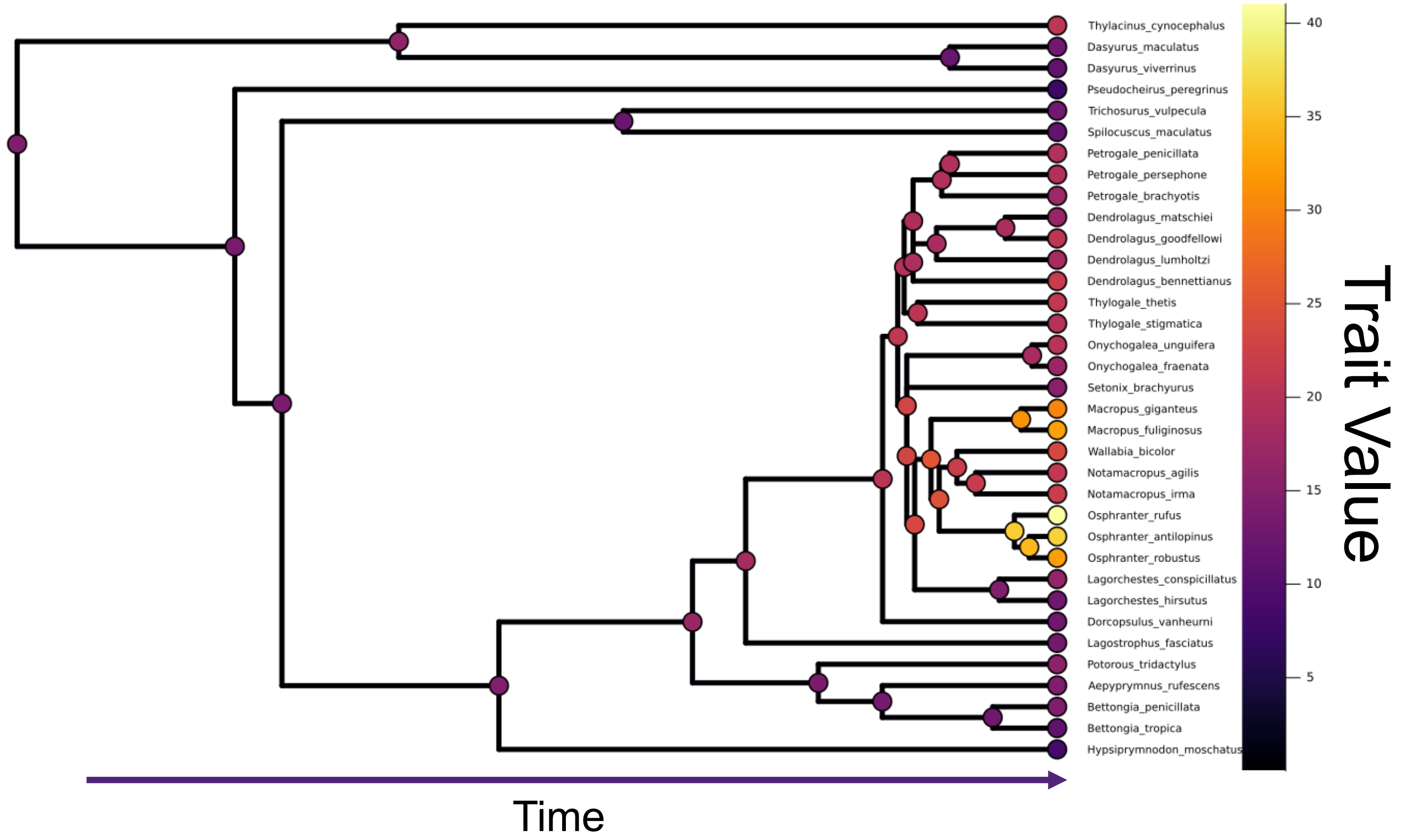


Figure 1: Ancestral trait reconstruction of 33 marsupial species for femur trochanteric fossa length value from 25 multiple imputations, from 25 independent initial samples with trait reconstruction repeated 25 times indicates the mean with heat map over the marker indicating predicted trait values, branch lengths show relative time

Bridge Density between *D. maculatus* and *D. viverrinus*

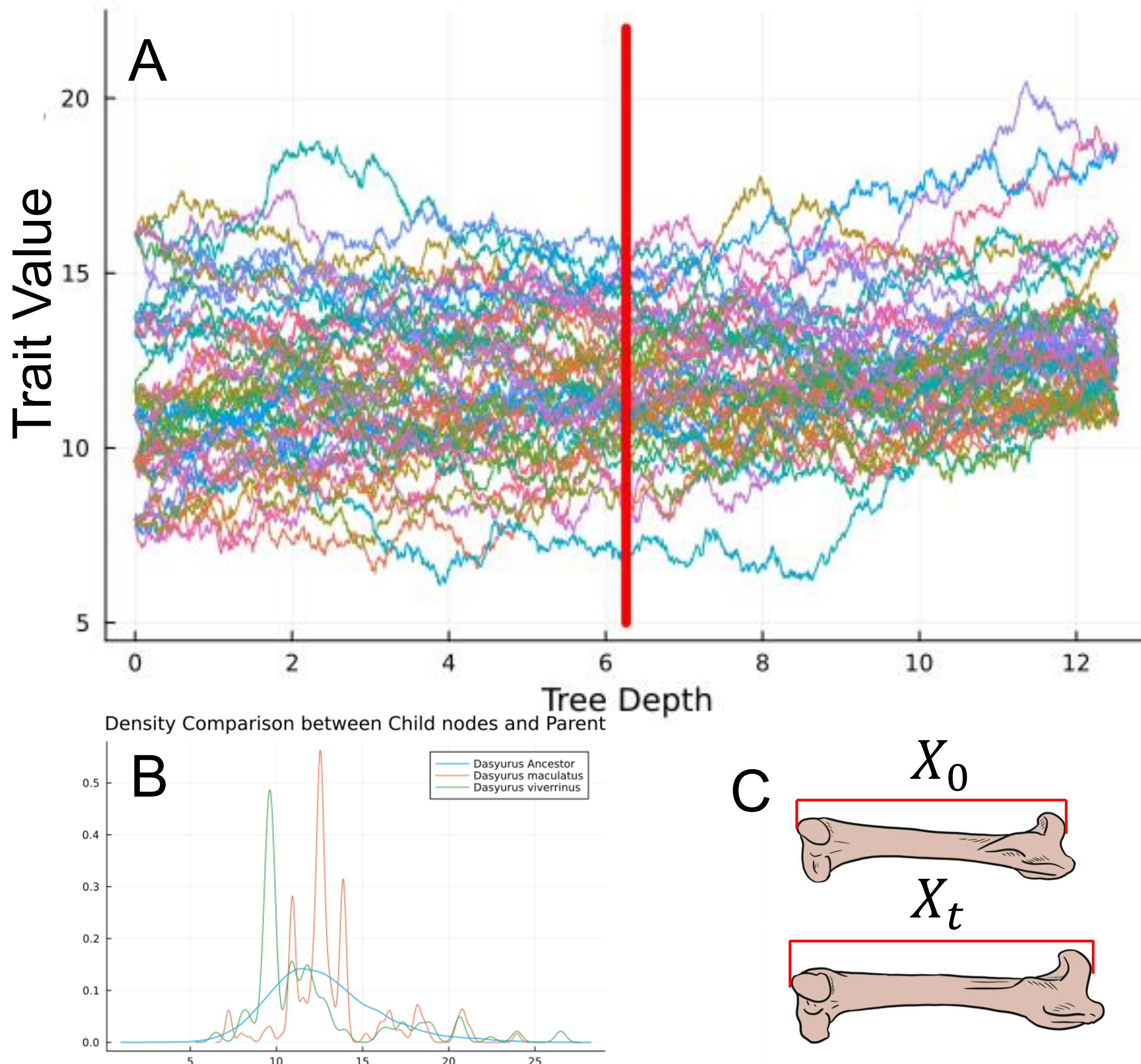


Figure 2: Brownian Bridge trait reconstruction between *Dasyurus maculatus* and *Dasyurus viverrinus*. **A)** contributing Brownian Bridges to the ancestral trait value of the ancestor, red line indicates the time point of the ancestor **B)** kernel density estimates of the contributing leaves to the resulting ancestral nodes, blue line corresponds to the density seen in figure A. **C)** measurements corresponding to initial densities of the bridges

References

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